

Luke Ferley

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Final-year Software Engineering student with experience building backend applications using Java, Spring Boot, Python, and AWS. Maintains an 83.4% academic average while mentoring peers in structured problem-solving. Passionate about clean code, Agile development, and Test-Driven Development, with a strong interest in building reliable, scalable software in collaborative engineering teams.

EDUCATION

WeThinkCode

Occupational Certificate in Software Engineering – 83.4% Year 1 Average

Johannesburg, Gauteng

Sept. 2025 – Expected Dec 2026

EXPERIENCE

Peer Tutor Volunteer

Oct 2025 – Present

WeThinkCode

Johannesburg, Gauteng

- Selected based on top academic performance to provide technical mentorship and support to fellow software engineering students.
- Tutored 12+ students in Java, Python, OOP, and data structures; assisted in refactoring legacy (brownfields) codebases and architecting new (greenfields) systems.
- Facilitated programming workshops and technical mock assessments, guiding peers through debugging and structured problem-solving to reinforce code quality.

Bootcamp Mentor Volunteer

Jul. 2026

WeThinkCode

Johannesburg, Gauteng

- Oversaw a group of five prospective students during a rigorous six-day selection bootcamp, tracking daily sprint progress, team engagement, and technical readiness.
- Mentored a team of six through Python fundamentals, collaborative version control, and algorithmic problem-solving to ensure successful group project delivery.

PROJECTS

Habit-Tracker API | Java, Spring Boot, PostgreSQL | [GitHub](#) | [API Docs](#)

May 2026 – Present

- Built a stateless REST API for habit tracking using Spring Boot and PostgreSQL.
- Secured user accounts with JWT authentication, Spring Security, and BCrypt password hashing.
- Developed a JDBC-based data access layer with parameterized SQL queries and Flyway-managed schema migrations, improving maintainability while reducing SQL injection risk.
- Validated application reliability through integration tests, Docker-based PostgreSQL environments, and API smoke testing with Insomnia.

Serverless Notes API | Python, AWS Lambda, DynamoDB | [GitHub](#) | [OpenAPI Spec](#) | [Demo](#)

Jul. 2026 – Present

- Architected and deployed a serverless REST API for note management using AWS Lambda, API Gateway, and DynamoDB, achieving 95% code coverage through automated testing.
- Developed Lambda handlers with boto3 and built an automated unit-testing suite with pytest and moto to improve deployment confidence.
- Produced an OpenAPI 3.0 specification to support API adoption and third-party testing.
- Automated deployments via GitHub Actions using an IAM OIDC policy for keyless authentication to AWS.
- Implemented CloudWatch monitoring and API smoke testing with Postman and Insomnia to improve operational visibility.

TECHNICAL SKILLS

Languages: Java, Python, SQL

Backend & Cloud: Spring Boot, AWS Lambda, API Gateway, DynamoDB, PostgreSQL

Developer Tools: Git, GitHub Actions, Docker, Postman

Practices: REST APIs, CI/CD, Unit & Integration Testing, Agile Development

CERTIFICATIONS & EXTRA LEARNING

CS50P: [Introduction to Programming with Python](#), HarvardX

AWS Educate: [Getting Started with Storage](#), [Introduction to Cloud 101](#), [Introduction to Generative AI](#)

Microsoft: [AI Fluency](#)